Total No. of Questions : 8]

SEAT No. :

P-1459

[Total No. of Pages : 2

[6004]-711

B.E. (Semester - VIII)

HONORS IN ELECTRIC VEHICLES

e-Vehicle standards, Charging and Safety

(2019 Pattern) (302036MJ)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right side indicate full marks.*
4) *Use of Calculator is allowed.*
5) *Assume Suitable data if necessary.*

Q1) a) List and explain in short any four battery charging and discharging parameters. **[9]**

b) How do you evaluate battery performance? Write any four methods to extend EV battery life? **[8]**

OR

Q2) a) Write a note on Control architecture in EV charging. **[9]**

b) Explain Electric Vehicle Charging and grid integration standards with any two suitable examples? **[8]**

Q3) a) Explain Testing procedures for following parameters of EV batteries : **[9]**

i) Battery performance

ii) Battery life

iii) battery safety

b) Write a note on Future trends in battery testing. **[9]**

OR

P.T.O.

Q4) a) Explain Specific Requirements for L, M and N Category Electric Power Train Vehicles with protection against direct and indirect contacts. **[9]**

b) How batteries of EV are tested? Explain mechanical and electrical testing for EV batteries **[9]**

Q5) a) Write a note on various failure modes and mechanism of Battery. **[9]**

b) Discuss in short the Safety and Abuse Response for Lithium Ion Rechargeable Battery Chemistries? **[8]**

OR

Q6) a) Explain the effects of following parameters on Li-ion batteries? **[9]**

i) Anode material.

ii) Cathode material.

iii) SOC on Thermal Stability.

b) Explain Evaluation Techniques for Batteries and Battery Materials. **[8]**

Q7) a) Write a note on EV Charging Infrastructure in India. **[9]**

b) Explain categories of charging stations. **[9]**

OR

Q8) a) Write a note on Standardization and regulations for EV Charging in india. **[9]**

b) Explain following impacts on power system **[9]**

i) Harmonic Impact.

ii) Harmonic Compensation.

iii) Current Demand Impact.

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